Response to Collaborator Goals

pmsimstats team

2026-03-22

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Date: 2026-03-23
Context: Following the pmsimstats code audit and revision

1 Code Locations

All new functions are in the `orig` repo at `~/Dropbox/prj/alz/01c-pmsimstats-orig/pmsimstats/`:

Function	File	Purpose
`analyze_trial_extended()`	`R/carryover_analysis.R`	Main drug effect + interaction + predicted curves
`characterize_carryover()`	`R/carryover_analysis.R`	Sweep half-lives, compare AIC, find best fit
`print_trial_summary()`	`R/carryover_analysis.R`	Clinical summary of trial analysis
`print_carryover_summary()`	`R/carryover_analysis.R`	Summary of carryover characterization
`lme_analysis()`	`R/lme_analysis.R`	Core LME analysis (nlme + corCAR1)
`generateData()`	`R/generateData.R`	Data generation with cached sigma
`buildSigma()`	`R/generateData.R`	Covariance matrix construction
`validateParameterGrid()`	`R/generateData.R`	PD pre-validation
`generateSimulatedResults()`	`R/generateSimulatedResults.R`	Full simulation loop with parallel processing

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2 Goal 1: Did prazosin improve PTSD symptoms?

How to analyze this in this trial design? – RCH implement with RGT guidance

2.1 What we have

The revised `lme_analysis()` uses `nlme::lme` with `corCAR1` (continuous-time AR(1) residual correlation), which is directly applicable to actual trial data. Vignette 3 in the original package demonstrates applying `lme_analysis()` to real clinical trial data (`CTdata.rda`). The model formula for the N-of-1 design is:

```
Sx ~ bm + t + Dbc + bm:Dbc, random = ~1|ptID,
  correlation = corCAR1(form = ~t|ptID)
```

2.2 What is needed

The current model tests the biomarker-treatment *interaction* (`bm:Dbc`): does blood pressure predict who responds? To test whether prazosin improves symptoms *overall*, the coefficient of interest is `Dbc` (the main drug effect), not `bm:Dbc`. This is a simpler test – check whether the drug indicator is significant.

The existing infrastructure supports this; it only requires extracting a different coefficient from the model output.

2.3 Action – DONE

Implemented in `R/carryover_analysis.R` as `analyze_trial_extended()`. Extracts both the main drug effect (`Dbc`) and the biomarker interaction (`bm:Dbc`) with 95% CIs, variance components, predicted response curves, and clinical decision thresholds.

Usage:

```r
result <- analyze_trial_extended(td, dat, op, threshold = 5)
print_trial_summary(result)
```

---

3 Goal 2: Characterize carryover in actual data

Need to figure out how to structure the carryover quantification and identify the best sequence of response measures. – RCH implement with RGT guidance

3.1 What we have

- The `lambda_cor = ln(2)/t_half` derivation provides a principled framework for relating carryover half-life to the correlation structure (see `pdf/08-biomarker-correlation-decay.pdf`).
- The `Dbc` continuous drug indicator with exponential decay is already implemented and parameterized by `carryover_t1half`.
- The `nlme::lme` with `corCAR1` analysis model handles the AR(1) residual structure.

3.2 What is needed

3.2.1 2a. Structure the carryover quantification

Fit the actual trial data with multiple carryover models (different half-lives) and compare fit via AIC/BIC or likelihood ratio test:

```
results <- list()
for (t_half in c(0.25, 0.5, 1.0, 2.0, 4.0)) {
  # Compute Dbc with this half-life
  # Fit nlme::lme with corCAR1
  # Record AIC, BIC, log-likelihood
  results[[as.character(t_half)]] <- list(
    t_half = t_half,
    aic = AIC(fit),
    bic = BIC(fit),
    loglik = logLik(fit)
  )
}
# Select best-fitting half-life
```

This is a straightforward extension of the existing `lme_analysis()`: loop over candidate half-lives and compare model fit. The `Dbc` computation is already parameterized by `carryover_t1half`.

A new function `characterize_carryover()` would:

1. Accept the trial data and design specification
2. Sweep a range of candidate half-lives
3. Fit the LME model at each half-life
4. Return a comparison table (half-life, AIC, BIC, log-likelihood, Dbc coefficient, p-value)
5. Identify the best-fitting half-life

3.2.2 2b. Identify best sequence of response measures

This likely means: at which timepoints does the carryover effect manifest most strongly? The existing trial design structure (with `tod`, `tsd`, `tpb` at each timepoint) provides the timing information. The analysis would:

1. Fit the model with the best carryover half-life
2. Extract per-timepoint residuals
3. Examine whether residuals at off-drug timepoints systematically differ from expectations (indicating carryover signal)
4. Plot the residual pattern across timepoints to visualize where carryover is strongest

3.3 Action – DONE

Implemented in `R/carryover_analysis.R` as `characterize_carryover()`. Sweeps candidate half-lives, fits nlme::lme at each, compares AIC/BIC, and returns the best-fitting half-life with full model output.

Usage:

```
cc <- characterize_carryover(td, dat,
  half_lives = c(0.25, 0.5, 1.0, 2.0, 4.0))
print_carryover_summary(cc)
# cc$best_t_half gives the optimal half-life
# cc$best_model gives the full model fit
```

Validated: correctly recovers the true t_half = 1.0 from simulated data generated with that half-life.

---

4 Goal 3: Analysis methods to add carryover

4.1 3a. Simulation power calculations

Done, partially; perhaps needs to be more modular/flexible. – RGT implement

4.1.1 Status: Essentially complete

The revised simulation code provides:

- `lambda_cor` parameter for correlation decay (auto-derived from half-life or manual override for sensitivity analysis)
- Carryover half-life as a simulation parameter: `c(0, 0.5, 1.0)` weeks (clinically realistic values replacing the publication's `c(0, 0.1, 0.2)`)
- `Dbc` continuous drug indicator with exponential decay in the analysis model
- 1,000-replicate validated power estimates with proper Type I error control (2.5–5.8% across all designs)
- 8-minute runtime (200 reps) enabling rapid sensitivity analysis; 47 minutes for 1,000 reps
- Zero positive definiteness failures across all 162 parameter combinations

4.1.2 Remaining modularity/flexibility items

1. **Carryover decay model options.** Currently exponential only. The `pmsimstats2025` repo has linear and Weibull decay in `calculate_carryover()`. These could be backported to allow sensitivity analysis over decay shape.
2. **Separate correlation decay rate.** The `lambda_cor` parameter currently defaults to `ln(2)/t_half` (matching the drug elimination rate). Making it an explicit sensitivity parameter allows exploring scenarios where biomarker predictive power decays faster or slower than the drug effect.
3. **With/without carryover in analysis model.** The simulation could compare power under two analysis scenarios: (a) carryover modeled in the analysis (continuous Dbc), and (b) carryover ignored (binary Db). This addresses the question: how much power is gained by correctly modeling carryover in the analysis?
4. **Modular design specification.** The trial designs are currently hardcoded in the simulation scripts. A configuration file or function that generates design specifications from high-level parameters (number of phases, block durations, randomization points) would make it easier to explore alternative designs.

4.2 3b. Actual analysis

In progress, reviewed 2/14 how to get the information needed in the output structure. – RGT implement

4.2.1 What we have

The `lme_analysis()` with `full_model_out = TRUE` returns:

- `form`: the model formula
- `fit`: the full `nlme::lme` model object
- `data`: the analysis-ready data with all derived variables (`Dbc`, `tsd`, `De`, etc.)
- `summary`: the standard output (beta, betaSE, p-value)

The full model object provides access to:

- `summary(fit)$tTable`: all coefficients with SEs and p-values
- `fitted(fit)`: predicted values per observation
- `residuals(fit)`: residuals per observation
- `ranef(fit)`: random effects (per-participant intercepts)
- `intervals(fit)`: confidence intervals for all parameters
- `VarCorr(fit)`: variance components (random intercept variance, residual variance, AR(1) parameter)

4.2.2 What is needed for the output structure

1. **Estimated carryover half-life** from the model comparison in Goal 2 (the best-fitting `t_half`).
2. **Predicted response curves per participant** using the fitted model: `predict(fit, newdata = ...)` with participant-specific random effects.
3. **Biomarker-specific treatment effect as a continuous function of blood pressure.** The `bm:Dbc` coefficient gives the slope: for each unit increase in blood pressure, the drug effect changes by β_4 CAPS points. Combined with the main `Dbc` effect, the predicted drug effect for a participant with blood pressure `bm` is: `effect(bm) = beta_Dbc + beta_bm:Dbc * (bm - mean_bm)`
4. **Clinical decision threshold.** At what blood pressure value does the predicted benefit exceed a clinically meaningful threshold (e.g., 5-point CAPS improvement)? This is: `bm_threshold = mean_bm + (threshold - beta_Dbc) / beta_bm:Dbc`

4.3 Action

DONE. Implemented as `analyze_trial_extended()` in `R/carryover_analysis.R`. Items 1–4 are all included: estimated carryover half-life (via `characterize_carryover`), predicted response curves (`result$predicted_effect(bm)`), biomarker-specific treatment effect with CIs, and clinical decision threshold (`result$threshold_bm`).

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5 Summary of Actions

Goal	Status	Action	Location
1. Prazosin efficacy	**DONE**	`analyze_trial_extended()`	`R/carryover_analysis.R`
2a. Carryover quantification	**DONE**	`characterize_carryover()`	`R/carryover_analysis.R`
2b. Best response sequence	Conceptual	Residual analysis at each timepoint	Needs implementation
3a. Simulation power	**DONE**	Full simulation with optimizations	`R/generateSimulatedResults.R`
3b. Actual analysis output	**DONE**	Extended output with predictions	`R/carryover_analysis.R`

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6 Key Deliverables from the Audit

The following documents and code are available:

6.1 White Papers

- `pdf/04-revised-power-analysis.pdf` (13 pages): Complete revision summary for the authors, with before/after Figure 4 heatmaps and all code changes

- `pdf/18-response-parameter-sensitivity.pdf` (6 pages): Response parameter sensitivity analysis
- `pdf/08-biomarker-correlation-decay.pdf` (10 pages): Pharmacokinetic derivation of the correlation decay rule
- `pdf/06-ar1-residual-correlation.pdf` (8 pages): Analysis model transition from lmer to nlme with corCAR1
- `pdf/07-positive-definiteness-failures.pdf` (10 pages): Positive definiteness failure analysis
- `13-figure4-walkthrough.md`: Complete technical walkthrough of Figure 4 methodology and code

6.2 Simulation Scripts

- `analysis/figure4/04-generate-power-head.R`: Figure 4 simulation (current code)
- `analysis/figure4/01-generate-power-sweep.R`: Figure 4 power sweep (sequential)
- `analysis/figure4/02-generate-power-sweep-original.R`: Publication reproduction
- `analysis/figure4/03-generate-power-by-commit.R`: Per-commit comparison
- `analysis/figure4/05-pd-diagnostics.R`: PD failure instrumentation
- `analysis/figure4/06-plot-power-heatmaps.R`: Heatmap plotting
- `analysis/figure5/01-generate-parameter-sensitivity.R`: Figure 5 simulation (revised)
- `analysis/figure5/02-generate-parameter-sensitivity-original.R`: Figure 5 (original)

6.3 Key Results

- 1,000-replicate Figure 4 with zero PD failures
- Type I error: 2.5–5.8% (all nominal)
- N-of-1 carryover sensitivity confirmed at realistic half-lives (0.5–1.0 weeks)
- Design ranking preserved: N-of-1 > CO > OL+BDC > OL
- Runtime: 8 minutes (200 reps), 47 minutes (1,000 reps)